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Multiobjective optimization for sizing and placing electric vehicle charging stations considering comprehensive uncertainties

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Abstract
The rapid growth of electric vehicles (EVs) demands a robust and efficient charging infrastructure. To address this, we propose a particle swarm optimization algorithm designed for optimal placement and sizing of EV charging stations. This study hypothesizes that comprehensive consideration of uncertainties in vehicle types, user behaviors, road dynamics, and environmental impacts will enhance infrastructure effectiveness. Our method integrates data from road networks, driver patterns, station owners, and EV manufacturers to meet diverse charging requirements. Results indicate that 14 fast charging stations are needed along the studied freeway, with a total installation cost of $289,820 and annual operational costs of $4,223,050, leading to annual CO_2 emissions of 1,843,572.57 kg. This strategic approach balances technical, environmental, and economic criteria, providing an essential tool for policymakers and urban planners in establishing sustainable EV charging networks.

Keywords Charging station planning, Particle swarm optimization, Environmental impact, Real-world uncertainties, Sustainable mobility

Introduction

In the dynamic landscape of advancing electric mobility, optimizing the complexities of electric vehicle (EV) charging infrastructure is imperative. The uncertainties associated with various factors, such as EV characteristics, travel parameters, station-related costs, and environmental impacts, present significant challenges. Addressing these uncertainties comprehensively is crucial for the efficient and sustainable development of EV charging networks [1].

Existing studies have made significant contributions to understanding these complexities but also reveal areas requiring further exploration. For instance, [2] delves into EV-specific uncertainties, focusing on travel distance, starting time, and geographic information, aligning to integrate these factors comprehensively. However, explicit consideration of battery capacity and state-of-charge uncertainties, which are crucial for a complete understanding of EV performance, is lacking.

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In [3], uncertainties in the types of EVs and travel parameters were considered to complement the multiobjective optimization methodology. However, it does not comprehensively cover the range of uncertainties related to installation cost, operation cost, and CO_2 emissions, leaving gaps in understanding the full economic and environmental impact.

Similarly, [4] provides insights into uncertainties related to EV characteristics and travel parameters but also overlooks installation costs, operation costs, and CO_2 emissions [5] explored uncertainties in the types of EVs, travel distance, starting time, and utilization factors via a comprehensive approach. Nonetheless, detailed examinations of waiting time and travel finishing time uncertainties, which are critical for user satisfaction and operational efficiency, are lacking.

Moreover, [6] presented a holistic approach considering EV characteristics, travel parameters, and environmental impact, which was combined with a multiobjective optimization methodology. Despite its comprehensive nature, a detailed examination of the waiting time, utilization factor, and travel finishing time uncertainties may be lacking. In [7], uncertainties related to the types of EVs, travel distance, starting time, and utilization factors contributed valuable insights to the research. However, this approach does not provide a comprehensive view of the installation cost, operation cost, or CO_2 emissions, which are essential for economic sustainability.

[8] explored uncertainties in the types of EVs, travel distance, and utilization factors, providing holistic insights. Nevertheless, it does not offer a comprehensive view of installation cost, operation cost, or CO_2 emissions. Additionally, [9] addresses uncertainties in travel parameters and installation costs, contributing to a broader understanding, but does not delve into waiting time, utilization factor, and travel finishing time uncertainties.

The studies reviewed provide a comprehensive overview of various strategies for optimizing EV charging infrastructure while addressing grid impacts and system reliability. In [10], the effects of different EV charging methods are compared based on peak demand load and voltage deviations, also considering renewable resources like wind and identifying off-peak hours as the optimal time for charging. In [11], a multi-objective framework models the placement of CS to ensure cost-effective grid operation, utilizing a chicken swarm optimization heuristic to analyze uncoordinated, coordinated, and bidirectional charging scenarios. In [12], a smart parking system employing autonomous learning algorithms manages EV charging efficiently, showcasing the benefits of digitized energy management without stressing the grid.

Further insights come from studies like [13], which detail Indian charging infrastructure planning, emphasizing factors such as demand-side management and geographic conditions but omitting grid considerations. In [14], the integration of DG units with EV charging stations is optimized using HGWOPSO, an AI method validated on IEEE-33 and IEEE-69 bus systems, highlighting improvements in grid reliability through strategic DG and EV integration. The rapid expansion of India's EV market is also explored in [15], examining charging specifications and regulatory guidelines while assessing future infrastructure plans. Lastly, [16] presents a practical method for sitting FCSs in the east delta network, incorporating photovoltaic distributed generation to mitigate real and reactive power losses and enhance system reliability, though voltage fluctuations remain unaddressed.

Finally, [17] provides insights into uncertainties related to the types of EVs, battery capacity, and travel distance. However, there are gaps in considering the state of charge, travel starting time, and geographic information uncertainties, which are essential for optimizing the charging infrastructure and achieving daily load forecasts for slow and fast charging but do not account for the uncertainty of the initial state of charge or provide load forecasts for each station individually. The main features of previous related studies are summarized in Table 1.

However, there are notable gaps in the literature. For example, [18] forecasts a collective daily electrical load for a whole integrated station into a distribution system without considering the optimal placement of individual charging stations. In [19], an algorithm for the optimum sizing and location of an EVFCS based on a fixed initial state of charge was presented, and the station load was forecasted as an average constant power rather than an hourly curve load.

Informed by the nuanced foundation laid by previous studies, this research introduces a cutting-edge multiobjective optimization technique for sizing and placing charging stations. This technique comprehensively addresses uncertainties encompassing the types of EVs, battery capacity, state of charge, travel distance, travel starting time, geographic information, installation cost, operation cost, and CO_2 emissions. By leveraging a comprehensive dataset merging road, driver, station owner, and EV manufacturer data, the proposed refined approach optimally guides station placement and strategically determines the number of charger type combinations for four distinct EV battery types. As shown in Fig. 1, the proposed model ensures a tailored and resilient charging infrastructure that balances the technical and environmental criteria.

This approach utilizes Particle Swarm Optimization (PSO), a mature optimization technique known for its robustness in handling many variables and uncertainties. PSO simulates the social behavior of birds flocking or fish schooling to find optimal solutions in complex spaces. The benefits of using PSO in the proposed model are manifold. First, the flexibility of PSO allows it to handle multiple objectives simultaneously, such as minimizing installation and operational costs while maximizing user convenience and minimizing CO_2 emissions. Second, PSO is highly effective in addressing the nonlinear, multidimensional nature of the problem, making it suitable for optimizing EV charging infrastructure because of its numerous interdependent variables. Third, the convergence properties of PSO ensure that the algorithm can find high-quality solutions within a reasonable computational time, which is crucial for practical applications [20].

This integration of PSO builds upon insights gleaned from a diverse array of literature, advancing the optimization of charging infrastructure while paving the way for a personalized and sustainable future of electric mobility. By addressing a broader range of uncertainties and employing a sophisticated optimization technique, this research significantly enhances the robustness, efficiency, and environmental sustainability of EV charging networks.

Additionally, this paper addresses a significant gap in the literature by forecasting the daily electrical load profile after defining the optimal locations of road stations. By defining the individual load for each station, the proposed approach provides a more accurate and granular understanding of the behavior of the electrical distribution grid. This detailed load forecasting allows for better planning and management of the grid, ensuring that it can accommodate the varying demands of multiple charging stations without

Table 1 Main features of previous studies

Ref	Uncertainty							Objectives				
	Types of EV	Battery capacity	State of charge	travel distance	travel starting time	Geographic information	travel finishing time	Waiting time	Utilization factor	Installation cost	Operation cost	CO_2 emission
2						✓				✓	✓	✓
3						✓				✓	✓	
4						✓				✓	✓	
5						✓				✓	✓	✓
6						✓	✓	✓	✓	✓	✓	✓
7					✓	✓	✓	✓	✓	✓	✓	
8	✓	✓		✓	✓	✓		✓	✓	✓	✓	✓
9				✓		✓			✓	✓	✓	
10	✓	✓	✓	✓		✓						

overloading. This precision helps maintain grid stability, optimize energy distribution, and reduce the risk of blackouts or other grid-related issues, ultimately contributing to a more reliable and resilient power infrastructure.

Key contributions of this study include:

Comprehensive Uncertainty Analysis: This study considers a wide range of uncertainties, such as EV types, battery capacity, state of charge, travel distance, hourly traffic flow, geographic information, installation and operational costs, and CO_2 emissions, often overlooked in previous studies.

Unified Road Model with Multi-source Data Integration: The model integrates datasets from road networks, driver behaviors, station owner considerations, and EV manufacturers to ensure optimal station placement and sizing.

Station-specific Load Forecasting: By providing individual load forecasts for each charging station, this model offers a more accurate and granular understanding of grid behavior compared to prior studies, which forecast only for generalized or aggregated systems. This precision aids in grid stability, optimal energy distribution, and the reduction of overload risks.

Sustainable Infrastructure Optimization: The proposed model strategically guides station placement, determining the number and type of chargers required for various EV battery types to achieve a balance between technical efficiency and environmental impact.

The framework of the article, as shown in Fig. 2, begins with a literature survey on previous research related to the optimal sizing and placement of FCS. This is followed by an examination of the technical and economic considerations involved in the optimal sizing and placement process. Next, the methodology used is presented, followed by a prediction of the daily load profile for each station. The economic impact, including capital, operational, and environmental costs, is then analyzed. Finally, the article concludes with a summary and suggestions for future work.

The paper is structured as follows: Sect. "Technical Background" reviews the necessary technical background for FCS modeling. Sect. "Methodology" outlines the methodology of the proposed multiobjective optimization model. Sect. "Results and Discussion" presents a case study of Hainan Island, detailing the simulation parameters and results. Sect. "Conclusion" discusses the implications of the findings and the robustness of the proposed approach. Finally, Sect. 6 concludes the paper by summarizing key contributions and suggesting directions for future research.

Technical background

EV/battery types

Failure to consider diverse EV types and their corresponding charging needs can result in inadequate infrastructure, hindering accessibility and convenience for users. Neglecting variations in battery capacity may lead to insufficient charging solutions, causing longer waiting times and dissatisfaction among users with different battery capacities. Additionally, overlooking the state of charge can result in inefficient charging processes, leading to incomplete charges and unnecessary sessions, impacting both the user experience and overall energy efficiency.

Incorporating a comprehensive understanding of EV types is crucial for effective planning. An approach utilizing two parallel classifications based on transport use and

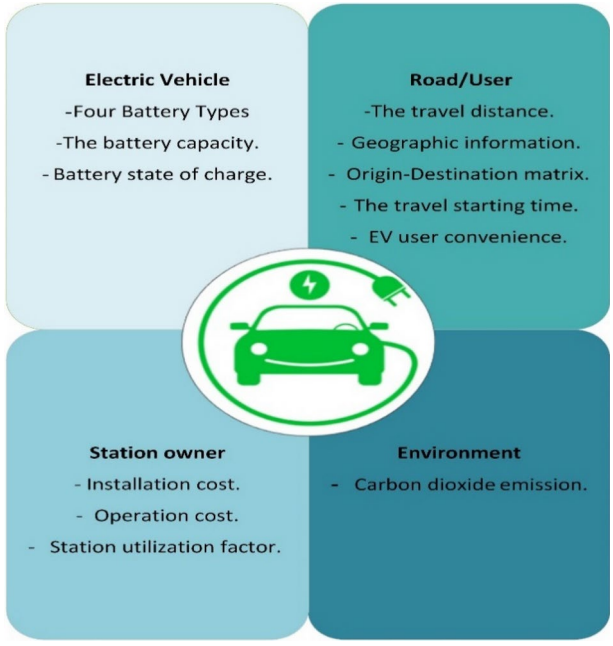


Fig. 1 The proposed model considerations

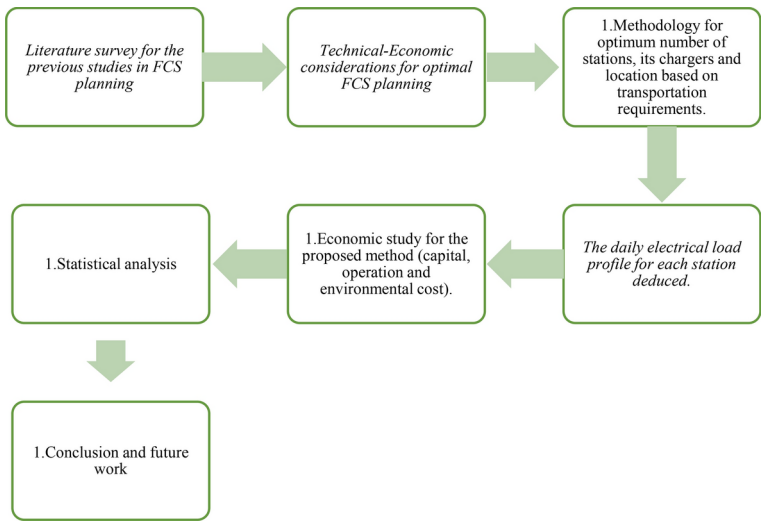


Fig. 2 Study framework

vehicle type enhances the spatial temporal model (STM). The classification considers EV travel patterns based on usage scenarios (Home-Based Work, Non-Home-Based, and Home-Based Others) [22] and identifies four vehicle types (L7e, M1, N1, and N2) [22]. Notably, the battery capacity, a key parameter, is modeled using probability density functions, ensuring realistic constraints for each EV type. Additionally, the real-time state of charge is dynamically calculated, considering factors such as travel distance and energy efficiency coefficients. This integrated approach aligns EV planning with real-world usage patterns and technical constraints, promoting a more accurate and efficient charging infrastructure.

According to the EV database, the probability density functions (pdfs) of the EV battery capacity (Cap) of different EV types (L7e, M1, N1, and N2) used in the Monte Carlo simulation (MCS) are shown in Eqs. (1) and (2). The parameters of each pdf are defined by Eq. (1) for the gamma distribution and Eq. (2) for the normal distribution [23]. For each EV, the MCS is used to generate Cap based on the corresponding pdf and constraints. If the capacity generated is outside the maximum (Max) and minimum (Min) kWh constraints, the process is repeated until the constraints are satisfied.

$$f\left(Cap, \alpha, \beta\right) = \frac{1}{\beta^{\alpha}\Gamma(\alpha)} Cap^{\alpha-1} e^{\frac{-Cap}{\beta}} \tag{1}$$

$$f(\mathrm{Cap}, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(Cap-\mu)^2}{2\sigma^2}} \tag{2}$$

where α and β are the shape parameter and scale parameter of the *pdf* of the gamma distribution, respectively, and μ and σ are the mean and standard deviation of the *pdf* of the normal distribution, respectively.

Road/user

Accurate consideration of travel distances in charging station planning is paramount. Insufficient infrastructure due to neglecting these distances, particularly in regions with longer routes, may leave EV users stranded, eroding confidence in the charging network's reliability. Additionally, overlooking the origin-destination (OD) matrix can lead to charging stations being misaligned with user demand, resulting in inefficient resource utilization and operational challenges. Neglecting hourly traffic flow may induce congestion during peak hours, causing longer waiting times, user dissatisfaction, and strain on the local power grid. Moreover, dismissing user convenience factors, including waiting areas and amenities, can significantly impact overall satisfaction. Therefore, a meticulous and user-centric approach to charging station planning is vital for providing optimal service and enhancing the electric vehicle user experience. In the following subsections, the travel distance, OD matrix, geographic information, and waiting time factor are considered.

Travel distance

The EV database provides the maximum travel range of an EV with its maximum battery capacity (Ran_{mc}) for different types of EVs. The mathematical relationship between Ran_{mc} and *Cap* is determined via polynomial fitting [24]. It is assumed that the *SOC* varies linearly with the real travel distance [25]-[26]. The travel range with available battery capacity (Ran_{ac}) of an EV is depicted in equation (3), and the EV travel range with the SOC_c (Ran_{sc}) of an EV is depicted in equation (4).

$$Ran_{sc} = \eta x \left(SOC_i - SOC_c\right) x Ran_{mc} \tag{3}$$

$$Ran_{sc} = \eta x \left(SOC_c\right) x Ran_{mc} \tag{4}$$

where η is an energy efficiency coefficient, which is introduced to consider the energy loss of an EV caused by the acceleration and deceleration processes. In this paper, it is assumed that SOC_i for an EV entering a road varies in the range of [0.2, 0.9].

Geographic information

The entrances and exits on the route are used as the origin and destination for OD analysis. It is assumed that EV users select the shortest path for their travels with an average velocity. The Floyd algorithm is used to determine the shortest path between the origin and destination locations [27].

Origin-destination (OD) matrix

Origin–destination (OD) analysis is used to model the spatial behavior of EV mobility, and transportation surveys are utilized to determine the temporal characteristics of EVs [26, 28, 29]. In urban areas, the daily travel pattern generally follows a regular pattern. An OD matrix $A_{NT} \times_{NT}$ is introduced in this study with probabilities (f_{OD}) assigned to each OD pair to represent the amount of traffic it attracts.

This value is generated using the gravity spatial interaction method [30], which suggests that the ability to attract the traffic flows between the OD pair depends on the weight assigned to the origin and the destination node and the distance between the OD pairs, as given in equation (5).

$$f_{OD} = \left\{ \begin{array}{c} 1.5x\frac{w_O w_D}{d_{OD}} if O \neq D \\ 0 if 0 = D \end{array} \right\} \tag{5}$$

where w_O and w_D are the weights assigned to the origin and destination nodes, respectively, and d_{OD} is the shortest route between the OD pair. Then, the traffic flow between the OD pair during the day at any time is determined in Eq. (6) using the average traffic that travels through the transportation network in time t. The information about the average traffic is obtained from the distribution of vehicles during the day given in transportation surveys.

$$f_{OD,t} = f_{total,t} x \frac{f_{OD}}{\sum_{O \in N_T} \sum_{D \in N_T} f_{OD}} t \in [1,24] \tag{6}$$

where $f_{total,t}$ is the average number of EVs that travel through the transportation network at time t and N_T is the set of transportation nodes. The generated EV mobility data for each OD pair are further used to estimate the number of EVs charging simultaneously at each FCS using queuing theory.

Waiting time

The charging of EVs can be a service provided by the FCS in a queue [31, 32]. The average number of EVs waiting in a queue for service is given by equation (7):

$$L_q^i = \frac{(\rho^i c^i)^{c^i+1} x P_0^i}{(c^i - 1)!(c^i - \rho^i c^i)^2} \tag{7}$$

where c is the number of chargers in the charging station and ρ is the station utilization factor.

In the above expressions, P_0^i denotes the probability that the CS is empty, with no EVs waiting for service, which is given by Eq. (8).

$$P_0^i = [\sum_{n=0}^{c-1} \frac{(\rho^i c^i)^n}{n!} + \frac{(\rho^i c^i)^c}{c^i!} x \frac{1}{(1-\rho^i)} \tag{8}$$

Therefore, the average waiting time for an EV user in the ith CS can be formulated as in equation (9):

$$W_t^i = \frac{L_q^i}{\lambda^i} \tag{9}$$

where λ^i is the parameter of the Poisson distribution representing the number of arriving EVs at the peak hour.

To ensure adequate service quality for EV users, the waiting time should not exceed the tolerable waiting time W_t^{max}, as expressed below in Eq. (10).

$$W_t^i \leq W_t^{max} i = 1,2,3,\ldots\ldots N^{CS} \tag{10}$$

Station owner

Neglecting installation costs in charging station planning can lead to significant financial challenges for station owners, hindering the growth of charging infrastructure and limiting accessibility for EV users. Similarly, disregarding operational costs may result in unsustainable charging station management, impacting the availability and reliability of the charging network. Additionally, overlooking the utilization factor of charging stations can lead to uneven resource distribution, with some stations being underutilized while others experiencing congestion, affecting overall system efficiency.

The initial investment required for charging station setup is a critical factor influencing the station economy. To design an optimum charging station location and size, minimizing installation costs is imperative. Installation costs are influenced by the number of fast chargers installed and may include fixed costs, land costs, and expenses for other auxiliary equipment. The fixed cost encompasses the basic machinery and equipment necessary for establishing a charging station, while land costs vary based on the chosen site.

One of the most economic objectives is to select the minimum number of chargers to reduce the installation and operation costs [6]. The number of required chargers is computed as in equation (11):

$$c_{min}^i = \frac{\lambda^i x c_{max}}{N_{EV}} \tag{11}$$

where λ^i is the arrival rate at the ith CS, c_{max} is the maximum allowable number of chargers, and N_{EV} is the total number of EVs considered for the FCS planning model.

The land cost for each station can vary depending on the site chosen for CS placement. In this work, we assumed that the installation of new EV charger equipment requires an area of 25 m^2 (A_c) with a clearance of 1 m between the chargers. The other auxiliary cost refers to the costs of essential equipment, such as distribution transformers and power cables, which are needed to set up a CS [6]. Therefore, the overall installation cost of one CS can be expressed as in Eq. (12).

$$C_{installation}^i = C_{fixed} + \left(A_c C_{land}^i + C_{ch} P_{rated}\right) c^i + C_o^i \tag{12}$$

C_{fixed}, C^i_{land}, C_{ch} and C^i_o denote the fixed cost, land cost, charger cost, and other auxiliary costs of the ith CS, respectively. Here, c^i represents the number of fast chargers equipped in CS i, and P_{rated} indicates the rated power of a fast charger.

The operational cost depends on the cost of electricity per unit and the power demand at each CS. The expected power demand at a CS can be evaluated by multiplying the rated power of the charger and the expected number of busy chargers, and the operational cost can be calculated as in equation (13) [6].

$$C^i_{operation} = C_{elec} x P_{rated} x \beta^i \tag{13}$$

where β^i denotes the expected number of busy chargers in CS 'i'.

Considering these cost factors and utilization parameters is essential for sustainable charging station planning, ensuring financial viability and optimal system efficiency. It is crucial to strike a balance between installation and operational costs while maximizing the utilization factor to create a financially sustainable and efficient charging infrastructure.

Another important criterion for capacity determination is the station utilization factor, which is defined as the proportion of time the station is busy. Therefore, each station should meet the waiting time constraint and station utilization constraint. The initial utilization factor (ρ_{ini}) is computed using the relation in Eq. (14) [6]:

$$\rho_{ini} = \frac{\lambda^i}{c^i_{min}\mu} \tag{14}$$

where μ denotes the charging service rate, i.e., the number of EVs charged in an hour.

Environment

Considering CO_2 emissions in EV charging station planning can jeopardize the sustainability of the charging network. Failing to adopt eco-friendly practices may contribute to increased carbon footprints, undermining the environmental benefits of electric vehicles. It is crucial to address these considerations collectively to create a robust, efficient, and environmentally friendly infrastructure that meets the diverse needs of EV users.

EVs, while having near-zero tailpipe emissions, still contribute to upstream emissions during the charging phase. This is mainly attributed to the electrical energy consumed, making it essential to consider CO_2 reduction as an objective in optimal charging station location decisions. The carbon emissions associated with EV charging depend on the fuel mix for electric power generation and the rate of carbon emissions per kilowatt hour [33].

Power plants fueled by fossil fuels emit more CO_2 than those fueled by renewable sources such as wind and solar energy. A mathematical formulation for estimating total CO_2 emissions considers factors such as the average emission factor.

A mathematical formulation for estimating the total CO_2 emissions of an EV can be computed as in equation (15).

$$CE_i = AEF x KWh \tag{15}$$

where *AEF* represents the average emission factor of the fuel mix, which is measured in grams of CO_2 emitted per kilowatt hour of electricity generated. KWh denotes the energy consumed for charging.

Incorporating these factors into charging station planning is vital for creating a sustainable infrastructure that aligns with environmental goals and enhances the overall positive impact of electric vehicles.

Individual station daily load curves

Once the charging station locations and the number of chargers at each station are determined using an appropriate algorithm, the next step is to predict the daily load curve of each station. The electrical daily load curve of a station is influenced by several factors, as outlined in the following subsections.

Individual daily traffic flow

The daily traffic flow at each station is represented by the number of vehicles arriving at each station at any given time. Historical data from the origin-destination (OD) matrix throughout the day are used to predict the daily traffic flow at each station.

EV charging time

The time required for charging an EV is determined based on its state of charge (SOC) and battery capacity when it arrives at the station. The charging time is calculated using equation (16) [18]:

$$T_{dc} = \frac{(1 - E_0)C_{battery}}{P_{charge} x \eta} \tag{16}$$

where $C_{battery}$ is the capacity of the EV battery and η is the charging efficiency, which mainly depends on the battery type.

The SOC of an EV upon arrival at the station is affected by the initial SOC when it enters the road (assumed to be a random value between 20% and 80%) and the distance traveled to the station. It is assumed that EV owners will recharge their vehicles as soon as the state of charge reaches the critical mode. Under this assumption, the SOC of the EV, denoted as $E_{0,}$ can be determined as depicted in equation (17) [18]:

$$E_0 = (SOC_{ini} - \frac{d}{L_{max}}) \tag{17}$$

where L_{max} is the maximum mileage of a fully recharged EV and d is a random variable indicating the total distance traveled to the station. From Eq. (17), it can also be concluded that the initial SOC of a single EV is also a random variable.

Station charger number

The optimal number of chargers at each station is calculated in the optimal capacity planning of the station. However, in practice, some chargers may experience faults. The number of operational chargers is a key factor in daily load forecasting. In this work, it is assumed that all chargers are in healthy mode.

Individual charger rated power

The EV charging power of a single EV is determined by the charging mode randomly selected by the EV drivers. In this work, it is assumed that the fast charging mode is applied. The rated power of the charger is 50 kW.

Methodology

The objective function of the planning algorithm is to minimize the costs associated with investment and operation and reduce the CO_2 emissions resulting from producing the energy required for charging. The following formula describes the objective function as in Eq. (18).

$$F = \min(w_1 \sum_{i=1}^{n} C_{inv,i} + w_2 \sum_{i=1}^{n} C_{op,i} + w_3 \sum_{i=1}^{n} CE_i) \tag{18}$$

where i denotes the station, n is the total number of stations and $\boldsymbol{w}_1, \boldsymbol{w}_2 \& \boldsymbol{w}_3$ is the weight of each value $\boldsymbol{w}_1 + \boldsymbol{w}_2 + \boldsymbol{w}_3 = 1$.

The number of charging locations (num) is first determined by (19).

$$num = fix(\frac{D_{od}}{Ran_{ac}}) \tag{19}$$

where the function 'fix' means rounding the value to the nearest integer toward zero. The locations of EV fast charging stations should satisfy the charging convenience of EV users. The convenience of EV users depends on factors such as the waiting time and the availability of charging stations along the road. As a result, a service radius (SR) is introduced to guarantee that each EV can drive to an adjacent charging station with an SOC_c.

The SR is determined based on the Ran_{sc} distribution supplied by the spatial–temporal model through the distribution of Ran_{sc} by using the normal fitting, as depicted in (20), according to the central limit theorem in probability theory [34].

$$\theta\left(Ran_{sc}, \mu_1, \sigma_1\right) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(Ran_{sc}-\mu_1)^2/2{\sigma_1}^2} \tag{20}$$

where μ_1 and σ_1 are the mean and standard deviation of the normal probability density function, respectively.

The SR is determined based on the normal distribution of Ran_{sc} with a (1-a%) confidence level, which is depicted in (21).

$$SR = \mu_1 - b\sigma_1 \tag{21}$$

To avoid redundant planning, the minimum distance between two adjacent EV charging stations (D_{acs}) follows the constraint depicted in (22).

$$D_{acs} \geq SR \tag{22}$$

To predict the daily load curve of each charging station, based on the transportation survey, the optimal location and size based on installation, operation and CO_2 emission costs should be determined.

According to gathered essential data on battery types and their characteristics, a Monte Carlo simulation (MCS) is used to simulate various scenarios for each battery

type to understand their performance under different conditions. A polynomial function is derived to establish the relationship between the maximum distance and battery capacity, which is crucial for determining the optimal station capacities. The geographical data of roads are collected to inform the positioning of charging stations. The number of stations required is calculated using equations that consider arrival rates from an OD matrix, and initial positions are randomly assigned.

By utilizing PSO, the minimum number of chargers is determined while meeting constraints such as the maximum waiting time and utilization factor. Iterative adjustments are made to the increment chargers if the constraints are not met, ensuring that the optimal capacity is achieved. The process evaluates costs and CO_2 emissions and employs a multiobjective function to assess each population member generated by the PSO algorithm. Iterations continue until the maximum criteria are reached, with the final output providing optimal station locations and capacities. Modifications may be necessary to clarify simulation parameters, refine constraint definitions, or optimize the parameters of the PSO algorithm for improved convergence and accuracy. All these arguments are illustrated in the flowchart in Fig. 3. Moreover, the pseudocode of the whole algorithm is explained in Fig. 4.

Results and discussion

Hainan Island's round-island freeway represents a unique case for the planning of an electric vehicle (EV) charging infrastructure. The freeway has 47 entrance/exit nodes [35], each of which plays a crucial role in the plan (see Fig. 5). The aim is to develop an efficient and convenient charging network that caters to current and projected EV traffic.

The distances between each entrance/exit and the reference entrance (location 1) are detailed in Table 2. The distances range from 0 km at the reference entrance to 587 km, reflecting the spread and accessibility of various points along the freeway. The total round-island distance is 612 km.

The average distance between each entrance/exit and the reference point is approximately 30.2 km. This average indicates a well-distributed network of points, which is crucial for ensuring accessible charging stations. The distances show a relatively uniform distribution with a slight increase in the middle segments, ensuring that no area is significantly underserved.

According to [20], 17,297 EVs will travel on a round-island freeway. The distributions of these EVs by type, i.e., L7e, M1, N1, and N2, are 10%, 84%, 3%, and 3%, respectively [21]. This distribution highlights the dominance of M1-type vehicles, which make up the bulk of the EV traffic. This information is critical for planning because it influences the types and quantities of chargers needed at each station. The simulation parameters are listed in Table 3, which are essential for accurately modeling EV charging needs and behaviors.

The probability density functions (PDFs) of the battery capacities for different EV types are summarized in Table 4 [14]. These functions are used in the Monte Carlo simulation (MCS) to predict the battery capacities of EVs and their charging requirements.

Figure 6 illustrates the relationship between the maximum travel distance, $R_{anmc,}$ and the battery capacity (Cap) for M1-type EVs [14]. Polynomial fitting provides a predictive model for understanding how different battery capacities impact travel distances,

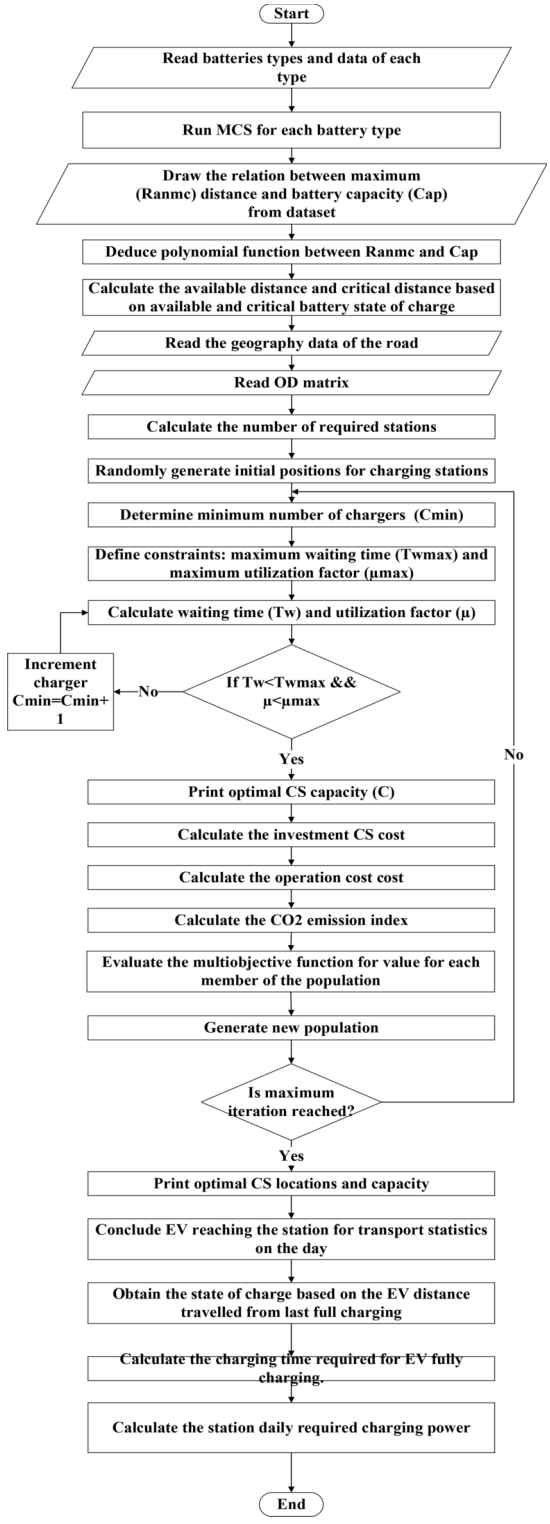


Fig. 3 Flowchart of the proposed algorithm

which is essential for strategically placing charging stations. Figure 7, extracted from the statistical data, shows the weights of the nodes throughout the day. These weights are used in the OD transport matrix simulation to understand traffic patterns and optimize

1. **Initialize:** Start the optimization process.
2. **Gather Battery Data:** Collect detailed characteristics and performance data for different battery types.
3. **Monte Carlo Simulation (MCS):** Simulate various scenarios using MCS to assess battery performance under different conditions.
4. **Derive Capacity-Range Relationship:** Establish a polynomial function to model the relationship between maximum range and battery capacity.
5. **Incorporate Geographical Data:** Integrate geographical information to influence the positioning and capacity calculation of charging stations.
6. **Calculate Demand and Initial Positions:** Determine charging station requirements based on arrival rates from an Origin-Destination matrix. Randomly initialize initial station positions.
7. **Optimization with Particle Swarm Optimization (PSO):**
   a. Initialize PSO parameters: Set up particle swarm optimization parameters, including population size, maximum iterations, and convergence criteria.
   b. Define Objective Function: Formulate an objective function considering factors like cost, CO_2 emissions, waiting time, and utilization factor.
   c. Evaluate and Update Particles: Evaluate each particle (representing a potential charging station configuration) based on the objective function. Update particle positions using PSO dynamics.
   d. Convergence Check: Determine if termination criteria (e.g., maximum iterations or convergence) are met.
8. **Output Results:**
   a. *Print Optimal Station Configuration:* Output the optimal positions and capacities of charging stations that meet defined constraints.
   b. *Evaluate Costs and Emissions:* Calculate the total costs and CO2 emissions associated with the optimal solution.
9. **Modify as Needed:** Review and adjust parameters, constraints, or algorithms based on simulation results for potential improvements.

Fig. 4 Proposed algorithm

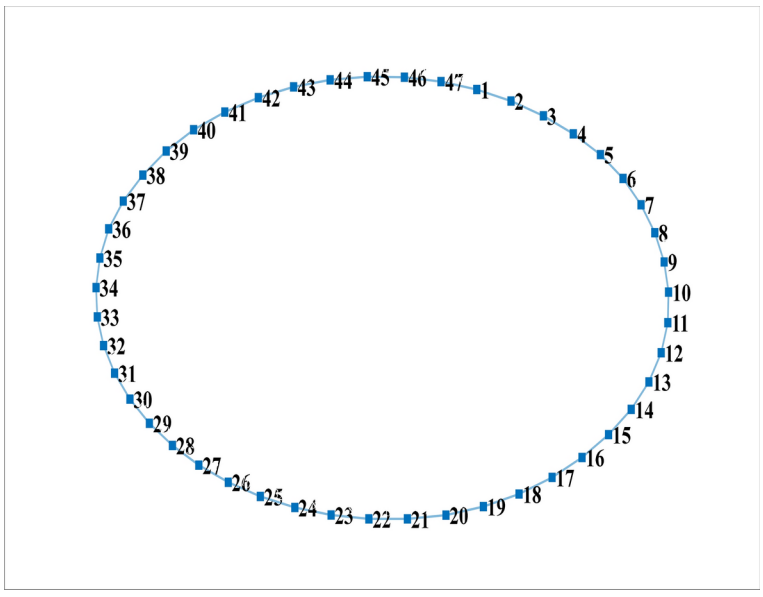


Fig. 5 The round-island freeway with entrances and exits

the placement of charging stations. High weights indicate peak travel times, which are critical for ensuring that charging stations can handle demand during these periods.

FCS size and allocation

The proposed algorithm for the optimal CS location and sizing was simulated using PSO in a MATLAB environment. The simulation results indicate the strategic placement of 14 FCSs along the studied freeway. The node numbers and distances, as well as the corresponding number of chargers at each location, are summarized in Table 5 and illustrated in Fig. 8.

The placement and sizing of the charging stations along the freeway are meticulously calculated to ensure optimal coverage, minimize waiting times, and support a balanced

Table 2 The entrances and exits of the test system

No	Distance(km)	No	Distance(km)	No	Distance(km)	No	Distance(km)
1	0	**13**	137	**25**	269	**37**	439
2	9	**14**	148	**26**	279	**38**	449
3	18	**15**	155	**27**	287	**39**	460
4	28	**16**	162	**28**	302	**40**	492
5	41	**17**	169	**29**	317	**41**	508
6	56	**18**	186	**30**	328	**42**	522
7	78	**19**	195	**31**	339	**43**	538
8	84	**20**	208	**32**	350	**44**	551
9	94	**21**	215	**33**	365	**45**	563
10	105	**22**	221	**34**	384	**46**	577
11	115	**23**	229	**35**	402	**47**	587
12	132	**24**	237	**36**	427	**Total**	612

Table 3 The simulation parameters

SOC_i	20%—80%
SOC_c	20%
η	0.95
b	1.65
a	5%
B_t	17,297
μ_2	6

Table 4 The PDF of the EV battery capacity in the MCS

EV group	L7e	M1	N1	N2
Distribution	Gamma	Gamma	Normal	Normal
Parameter	$\alpha=10.8$	$\alpha=4.5$	$\sigma=23$	$\sigma=85.3$
	$\beta=0.8$	$\beta=6.3$	$\mu=9.5$	$\mu=28.1$
Max (kw/h)	15	72	40	120
Min (kw/h)	5	10	9.6	51.2

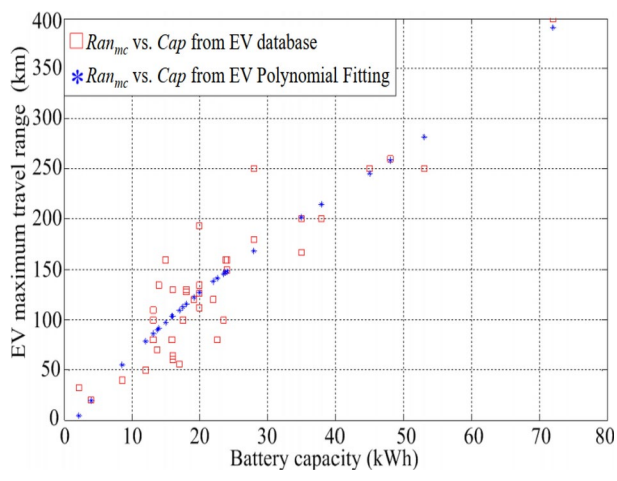


Fig. 6 Maximum travel distance/battery capacity relation

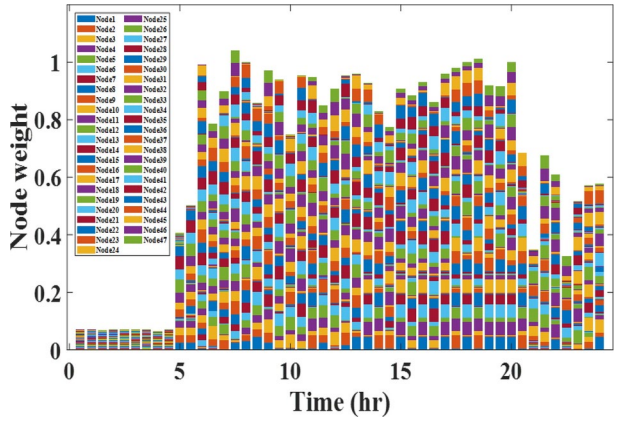
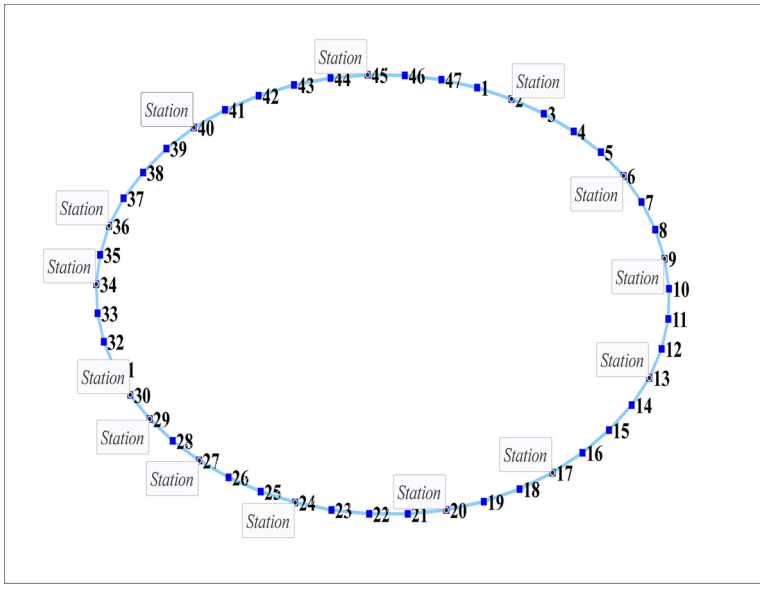


Fig. 7 The weights of nodes throughout the day

Table 5 Location and number of chargers in the FCS

Charging station number	Node number (distance)	Number of chargers
1	2 (9)	12
2	6 (56)	15
3	9 (94)	17
4	13 (137)	13
5	17 (169)	10
6	20 (208)	15
7	24 (237)	12
8	27 (287)	11
9	29 (317)	15
10	30 (350)	12
11	34 (384)	17
12	36 (448)	17
13	40 (522)	14
14	45 (563)	10


Station
Station
Station

load. Starting with Node 2 at 9 km, which is positioned close to the initial point of the freeway, it ensures early access for EV users beginning their journey amidst high traffic density. This station, equipped with 12 chargers, caters to the initial surge in charging demand. Moving to Node 6 at 56 km, an intermediate location, it serves as an essential stop for vehicles that use a significant portion of their charge, justifying its 15 chargers. Node 9 at 94 km marks a critical point almost a quarter of the way through the route, catering to long-distance travelers needing substantial recharge due to high expected demand, hence the 17 chargers.

Node 13 at 137 km helps distribute the load evenly across the network, ensuring that no station is overwhelmed by its 13 chargers. Strategically spaced node 17 at 169 km reduces range anxiety for EV users with moderate traffic flow and is adequately equipped with 10 chargers. Positioned beyond the halfway point, Node 20 at 208 km accommodates high demand from vehicles needing a second recharge with 15 chargers. Node 24 at 237 km has a balanced spacing, and steady traffic flow is managed with 12 chargers.

Node 27 at 287 km supports vehicles heading toward the next major stop, ensuring continuity in service with 11 chargers. Node 29 at 317 km provides critical support midway through the second half of the journey, sustaining moderate traffic with 15 chargers. Node 30 at 350 km anticipates high usage deep into the route, necessitating 12 chargers amidst a likely traffic surge. With 17 chargers, node 34, which is 384 km long, ensures that EVs can continue their travel without long distances between recharges. Node 36 at 448 km provides convenience and reduced waiting time with its 17 chargers.

Node 40 at 522 km accommodates battery types with less capacity, avoiding stops due to low charge by providing 14 chargers. Finally, node 45 at 563 km, near the endpoint, ensures that vehicles have sufficient charge to complete their journey with 10 chargers due to reduced demand as vehicles spread out.

This strategic distribution not only enhances user convenience and satisfaction but also promotes the efficient utilization of resources, reducing operational bottlenecks. Additional analyses, such as cost–benefit, user experience metric, environmental impact, sensitivity, and comparative analyses, further validate the algorithm's effectiveness and real-world applicability.

Economic/environmental feasibility

To evaluate the economic feasibility of the proposed method for the optimal location and sizing of FCSs, an economic analysis was conducted using the parameters listed in Table 6.

Using the values in Table 6 and the cost equations for installation, operation, and environmental impacts, the costs for each station were calculated, as shown in Table 7.

The total investment cost for installing the FCS network is \$289,820, with an annual operation cost of \$4,223,050, which includes only the cost of electricity. The annual CO_2

Table 6 The cost parameters

Fixed Station Cost C_{fixed} (\$/station)	1000
Land cost per unit area (\$/m^2)	15–55
Unit Charger Cost (\$/kW)	0.06
Auxiliary Cost per station C_0 (\$/station)	1000
Electricity price for charging (\$/kWh)	0.08
CO_2 emission factor of fuel mix (g CO_2/kwh) (AEF_{EV})	436.55

Table 7 The installation, operation cost and environmental impact

Station	Installation cost ($)	Operation annual cost ($)	Environmental annual impact (kg CO_2)
Station 1	18,536	299,300	130,659.4
Station 2	22,670	394,930	172,406.7
Station 3	25,426	218,270	95,285.77
Station 4	19,914	234,330	102,296.8
Station 5	15,780	316,090	137,989.1
Station 6	22,670	319,010	139,263.8
Station 7	18,536	353,320	154,241.8
Station 8	17,158	335,800	146,593.5
Station 9	22,670	232,870	101,659.4
Station 10	18,536	323,390	141,175.9
Station 11	25,426	278,130	121,417.7
Station 12	25,426	278,130	121,417.7
Station 13	21,292	323,390	141,175.9
Station 14	15,780	316,090	137,989.1
Total	289,820	4,223,050	1,843,572.57

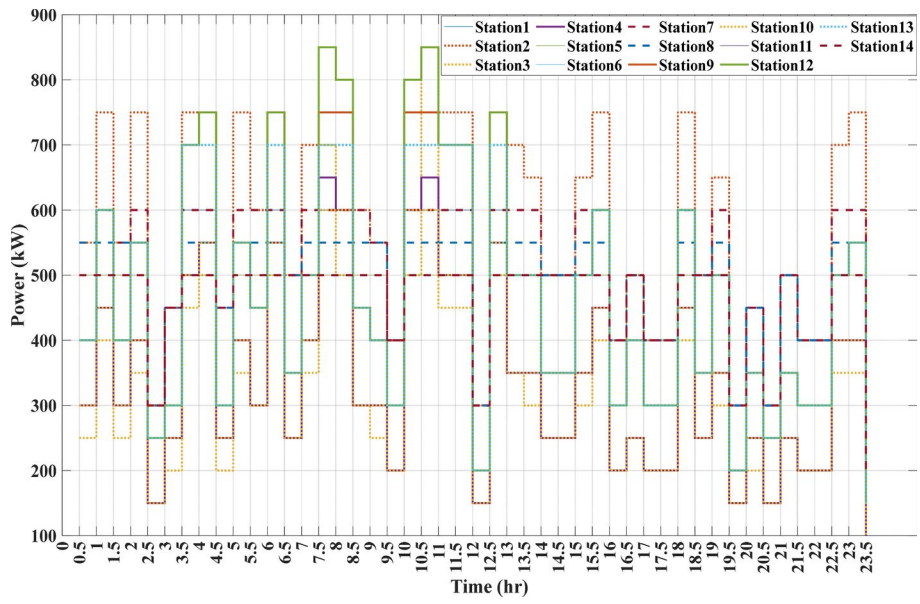


Fig. 9 The individual daily load curve of FCSs

emissions resulting from the energy required for charging are 1,843,572.57 kg of CO_2. The installation costs vary across stations and are influenced by the land cost per unit area and the number of chargers needed. The highest installation costs are observed at Stations 3, 11, and 12 due to their strategic locations and expected high traffic volumes.

The annual operation costs are significant, reflecting the extensive usage of electricity for charging EVs. For instance, Station 2 incurs the highest operational cost, which aligns with its strategic importance and high utilization rate. Stations 7 and 10 also exhibit high operational costs due to their busy locations and high demand for charging services.

The environmental impact, measured in terms of CO_2 emissions, is substantial across all stations. The CO_2 emission factor of 436.55 g CO_2/kWh contributes to these high annual emissions, highlighting the importance of integrating renewable energy sources to mitigate environmental impacts. Stations with the highest CO_2 emissions, such as

Table 8 Individual daily load curve statistics

Station	Mean load	Peak load	Load factor	Standard Deviation	Skewness	Kurtosis
CS 1	436.1702	600	0.056944	137.8019	-0.1995	1.7486
CS 2	575.5319	750	0.060111	165.7894	-0.4129	1.9174
CS 3	318.0851	850	0.029314	147.5987	1.3529	5.3938
CS 4	341.4894	650	0.041154	146.8211	0.4919	2.2427
CS 5	460.6383	500	0.072167	72.9325	-1.9041	5.8398
CS 6	464.8936	750	0.048556	178.4337	0.2955	1.9186
CS 7	514.8936	600	0.067222	109.7997	-1.0945	3.1672
CS 8	489.3617	550	0.069697	90.8512	-1.4601	4.2087
CS 9	464.8936	750	0.035444	178.4337	0.2955	1.9186
CS 10	339.3617	600	0.061528	142.5453	0.3993	2.0527
CS 11	471.2766	850	0.037353	189.8926	0.4538	2.1579
CS 12	471.2766	850	0.037353	189.8926	0.4538	2.1579
CS 13	457.4468	700	0.052738	166.8046	0.1599	1.8154
CS 14	460.6383	500	0.072167	72.9325	-1.9041	5.8398

Stations 2, 7, and 10, underscore the need for sustainable energy solutions to reduce the overall environmental footprint of the charging network.

Individual electrical daily load profiles

Using historical data on daily traffic flow [20], EV arrival rates at stations over time, the calculated state of charge (SOC) of each EV upon arrival, and the required charging duration, we predict the daily load curves for charging stations, as illustrated in Fig. 9.

The proposed algorithm plays a crucial role in forecasting the individual load profiles of each CS, which is essential for understanding the overall behavior of the power system. As shown in Table 8, The mean load across stations varied, with the highest mean load observed at CS 2 (575.5319 kW) and the lowest at CS 3 (318.0851 kW). The peak loads are notably high, with CS 3 and CS 11/12 reaching up to 850 kW, indicating significant demand at these stations. This underscores the importance of strategically placing and sizing charging stations to handle peak loads efficiently. The load factor, which indicates the efficiency of charging station utilization, ranges from 0.029314 (CS3) to 0.072167 (CS5 and CS14).

A higher load factor implies more consistent usage, while a lower load factor suggests variability and potential underutilization. The standard deviation values indicate the variability in the load profiles. Higher values, such as at CS 11/12 (189.8926), suggest greater fluctuations in demand, which the algorithm needs to account for in planning. The skewness values show asymmetry in the load distribution. For instance, CS 3 has a positive skewness (1.3529), indicating a longer tail on the right side of the distribution. Conversely, CS 5 has a negative skewness (-1.9041), showing a longer tail on the left side. Kurtosis values highlight the Preakness of the load distribution. Stations such as CS 3 and CS 5 exhibit high kurtosis (5.3938 and 5.8398, respectively), indicating sharp peaks and heavy tails, which can be challenging for load forecasting.

The ability of the proposed algorithm to forecast individual load profiles is pivotal for several reasons. By accurately predicting the mean and peak loads, the algorithm ensures that each charging station is appropriately sized, thereby avoiding both underutilization and overload. This leads to efficient resource allocation and better financial planning.

Understanding the variability and distribution of loads through metrics such as standard deviation, skewness, and kurtosis helps in planning for grid stability. The algorithm

can anticipate high-variability stations and implement strategies to mitigate potential disruptions. By minimizing waiting times and ensuring reliable availability of chargers, the algorithm enhances user satisfaction. This is achieved by strategically placing stations with the expected high demand and ensuring that they have the capacity to handle peak loads. Efficient load management and optimized station placement reduce unnecessary energy consumption and minimize CO_2 emissions. The algorithm supports the integration of renewable energy sources, aligning with sustainability goals. The algorithm provides detailed insights into load profiles, enabling better forecasting and long-term planning. This is crucial for adapting to future increases in EV adoption and ensuring that the infrastructure can scale accordingly.

The detailed analysis of load characteristics across different charging stations highlights the variability and demand patterns within the network. The ability of the proposed algorithm to accurately forecast these profiles is essential for optimizing station placement and sizing, ensuring efficient operation, and supporting the overall stability and sustainability of the power system. This strategic approach not only meets the current demand but also prepares the infrastructure for future growth, making it a vital tool for modern EV infrastructure planning.

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Conclusion

This study introduces an innovative and comprehensive methodology for optimizing the location and sizing of electric vehicle (EV) charging stations, using particle swarm optimization (PSO) to address the complexities and uncertainties inherent in integrating

plug-in electric vehicles (PEVs) into existing energy and mobility frameworks. By factoring in vehicle specifics, user behaviors, station costs, and environmental impacts, the algorithm identified a requirement of 14 fast charging stations along the studied freeway, with a total installation cost of $289,820 and annual operational costs of $4,223,050, contributing to CO_2 emissions of 1,843,572.57 kg per year. This balanced approach promotes user satisfaction, economic feasibility, and environmental sustainability.

Forecasting individual daily load profiles further demonstrates the algorithm's effectiveness in regulating power system dynamics, promoting load balance, and mitigating grid overload. This approach not only enhances user satisfaction and economic feasibility but also serves as a valuable tool for policymakers, urban planners, and stakeholders in developing sustainable EV charging networks.

Ultimately, this study supports sustainable transportation solutions, advancing carbon reduction efforts and fostering the adoption of clean energy technologies. The method's application on a single-direction ring road may not fully account for the complexities of bidirectional or multi-path road systems, which could limit its adaptability to more intricate urban or regional layouts. These limitations suggest that future work could expand the model to bidirectional and more complex road networks to improve the adaptability and robustness of the proposed method across diverse infrastructures also discuss the effect of the loads on the grid in absence and presence of renewable resources.

Abbreviations

EV	Electric vehicle
PSO	Particle swarm optimization
FCS	Fast charging stations
CO_2	Carbon dioxide
STM	Spatial–temporal model
PDF	Probability density function
MCS	Monto Carlo simulation
Cap	Battery capacity
Min	Minimum
Max	Maximum
OD	Origin–destination
Ran_{mc}	The maximum travel range of an EV with its maximum battery capacity
SOC	State of charge
SOC_c	Critical state of charge
SOC_i	Initial state of charge
Ran_{sc}	The travel range of an EV with its critical battery capacity
CS	Charging station
num	The number of charging locations
SR	Service radius
D_{acs}	The minimum distance between two adjacent EV charging stations
MCS	Monte Carlo simulation

Author contributions

AbdAllah Mohammed: Conceptualization, Acquisition, Data curation, Writing – original draft, Writing – review & editing, Visualization, Investigation, Validation, Formal analysis, Methodology, Supervision, Software. Omar Saif: Conceptualization, Acquisition, Data curation, Writing – original draft, Writing – review & editing, Visualization, Investigation, Validation, Formal analysis, Methodology, Supervision, Software. Maged A Abo Adma: Conceptualization, Acquisition, Data curation, Writing – original draft, Writing – review & editing, Visualization, Investigation, Validation, Formal analysis, Methodology, Supervision, Software. Rasha Elazab: Conceptualization, Acquisition, Data curation, Writing – original draft, Writing – review & editing, Visualization, Investigation, Validation, Formal analysis, Methodology, Supervision, Software.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests
The authors declare no competing interests.

Ethical approval
This study did not involve human participants, animals, or any form of biological material, and therefore, ethical approval was not required. The research conducted is based on engineering simulations and theoretical models. All data used in this study are publicly available or generated through validated computational methods, and no ethical concerns arose in the course of the work.

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